

Ernest van Wijland

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Education

- **PhD Computer Science** **IRIF, Université Paris-Cité**
Paris, France *2024 – Present*
 - PhD student under Claire Mathieu (CNRS) and Vincent Cohen-Addad's (Google Research) supervision.
- **MSc Computer Science, BSc Mathematics** **École Normale Supérieure**
Paris, France *2020 – 2024*
 - Graduated both the MSc (MPRI) and the BSc (double major in Mathematics and Computer Science) with highest honors.
- **Preparatory Class** **Lycée Louis-le-Grand**
Paris, France *2018 – 2020*

Publications

- **TCS-Bench: Benchmarking State-of-the-Art Generative AI TCS Research Ability** **In Submission**
*Cohen-Addad *, Paparas *, Van Wijland *, Springer, Canitrot-Paradis, Lin, Woodruff, Kumarappan, Jayaram, Das, Jain, Svensson, Lattanzi, Balunovic, Weber, Mirrokni (*: co-first authors)* *2026*
- **Beyond Worst-Case Online Multi-Level Aggregation** **In Submission**
Mathieu, Naor, Van Wijland *2026*
- **Spectral Dual Fitting for k -Means** **In Submission**
Anand, Charikar, Cohen-Addad, Gao, Grandoni, Lee, Sharma, Van Wijland *2026*
- **A $(4 + \varepsilon)$ -Approximation for Euclidean k -Means via Non-Monotone Dual Fitting** **STOC**
Charikar, Cohen-Addad, Gao, Grandoni, Lee, Van Wijland *2026*
- **An Improved Greedy Approximation for (Metric) k -Means** **FOCS**
Charikar, Cohen-Addad, Gao, Grandoni, Lee, Van Wijland *2025*
- **Faster Approximation Scheme for Euclidean k -TSP** **SOCG**
Van Wijland, Zhou *2024*
- **Maximizing Nash Social Welfare in 2-Value Instances: Delineating Tractability** **MOOR Journal**
Akrami, Chaudhury, Hoefer, Mehlhorn, Schmalhofer, Shahkarami, Varricchio, Vermande, Van Wijland *2023*
- **Maximizing Nash Social Welfare in 2-Value Instances** **AAAI**
Akrami, Chaudhury, Hoefer, Mehlhorn, Schmalhofer, Shahkarami, Varricchio, Vermande, Van Wijland *2022*

Experience

- **Student Researcher** **Google Research**
Paris, New York City *2026 – Present*

Currently working under Silvio Lattanzi and Vincent Cohen-Addad's supervision at Google Research on AI's mathematical abilities.
- **PhD Student: Clustering and Network Design** **IRIF (Université Paris-Cité)**
Paris, France *2024 – Present*

Currently working under Claire Mathieu and Vincent Cohen-Addad's supervision at IRIF on finding

approximation algorithms for clustering and network design problems.

- **Research Intern: Approximating TSP generalizations** **IRIF (Université Paris-Cité)**
Paris, France 2023 – 2024
Worked for a year under Claire Mathieu and Hang Zhou's supervision at IRIF on finding fast approximation schemes for generalizations of the TSP, in Euclidean spaces and general metric spaces.
- **Research Intern: Euclidean CVRP, k -TSP** **LIX (Polytechnique)**
Palaiseau, France 2023
Worked for a semester under Hang Zhou's supervision in Polytechnique on finding faster approximation schemes for the Euclidean CVRP and k -TSP.
- **Research Intern: Approximating 2-WECS** **THL (EPFL)**
Lausanne, Switzerland 2022
Worked for a semester under Ola Svensson's supervision in the Combinatorial Optimization team at EPFL on finding a better than 2 approximation for 2-WECS.
- **Research Intern: Fair Division for additive instances** **AGT (Max Planck Institut)**
Sarrebruck, Germany 2021
Worked for two months under Kurt Mehlhorn's supervision in the Algorithmic Game Theory team of the MPII on the Fair Division problem for additive instances.
- **Research Intern: Spectrum of heavy-tailed random matrices** **DMA (École Normale Supérieure)**
Paris, France 2021
Worked under Laure Dumaz' supervision on a paper by Ben Arous and Guionnet.

Academic achievements

- **Best French Developer Contest 5th place finisher**
Paris, France 2023
 - French national contest in Computer Science, with thousands of participants from every major IT company
 - awarded the 5000€ Genius prize as a high scoring student
 - **International Olympiad in Informatics Bronze medalist**
Tsukuba, Japan 2018
 - **Dutch National Olympiad in Informatics winner**
Utrecht, Netherlands 2018
 - **Innopolis University Olympiad in Informatics finalist**
Innopolis, Russia 2018
 - **International Olympiad in Informatics participant**
Tehran, Iran 2017
- As part of the top 4 High-School students in the Dutch National Olympiad in Informatics.

Outreach

- **Member of the France-IOI association**
France 2018 – Present
I am involved in organizing the Castor and Algoréa contests, in which more than 700 000 students from elementary school to high-school participate every year. I am also involved in selecting and training the best students that participate in the Informatics Olympiad.
- **Member of the Netherlands-IO association**
Netherlands 2018 – Present
I am involved with the computer science outreach campaign towards Dutch high-school girls and preparing the Dutch girls team for the European Girls Olympiad in Informatics.
- **Lecturer at the Moroccan Olympiad in Informatics training camp**
Morocco 2019

Gave algorithms lectures to Moroccan middle-school and high-school students.

Other interests

- Active Judo black belt and Tennis player.
- Practicing piano for the past 20 years, part of the chamber music club in the ENS. Founder of the Early Music section.
- Graduated from the Juvisy Music Conservatory in Music Theory, Piano and Ensemble Playing (2018).
- Graduated ENS with a Musicology Minor.